NORTHWEST INTERNATIONAL AIR QUALITY ENVIRONMENTAL SCIENCE AND TECHNOLOGY CONSORTIUM (NW-AIRQUEST) CHARTER

May 19, 2020

Introduction

At the 2003 Northwest Air Summit, nearly 200 participants from business, non-profit organizations, and local, state, tribal, and federal governments agreed that air quality management decisions should be based on sound science and proposed the establishment of the "Northwest International Air Quality Science Coordinating Committee." This decision built on past collaborations between air quality scientists in southwestern Canada and the northwestern United States. In December 2003, the group was formalized as the Northwest International Air Quality Environmental Science and Technology Consortium (NW-AIRQUEST). A charter dated December 4, 2003 was signed by participating organizations in the following months. The charter provided a foundation for enhanced communication of the science at both the technical staff and management levels of participating organizations, and the opportunity to collectively seek financial backing for projects of common interest.

NW-AIRQUEST operates as a parallel organization to the Northwest Regional Modeling Consortium (NRMC), which provides mesoscale meteorological forecasts for several related scientific disciplines. An organization's membership and membership category in NW-AIRQUEST is independent of its association with the NRMC and vice versa. However, NW-AIRQUEST depends heavily on NRMC products and NW-AIRQUEST's sustaining members are also typically members of NRMC, and therefore fund both NRMC products (e.g., WRF) and NW-AIRQUEST products (e.g., air quality forecasting products such as AIRPACT).

The original 2003 NW-AIRQUEST charter was updated in 2010. The update was motivated by the NRMC effort to expand its membership to include heavy users of its products. In that process, the air quality community saw a need to re-affirm and modify the role of NW-AIRQUEST and its membership categories.

Ten years later, NW-AIRQUEST and the NRMC are still actively working on meteorological and air quality products needed by their sustaining members. The 2010 NW-AIRQUEST charter update has expired, and therefore the charter is being renewed in 2020. This updated charter describes the purpose and goals of NW-AIRQUEST, and the framework within which it functions to achieve these goals. For the purposes of this charter, the region concerned is the Pacific Western North America Region taken to encompass all or part of the states of Washington, Oregon, Idaho, Montana, and Alaska in the United States, and the provinces of British Columbia and Alberta in Canada.

Purpose

NW-AIRQUEST seeks to maintain and enhance a sound scientific basis for air quality management decision-making in the Pacific Western North America Region. It achieves such a basis through collaboration and sharing of experiences to remain abreast of the state-of-the-science and to develop regionally consistent technical approaches to emission inventories, air quality modeling, and air monitoring. NW-AIRQUEST cooperation enhances the effectiveness of individual members in a mutually beneficial relationship while operating within the resource constraints of each member organization. NW-AIRQUEST collaborates to align science and management needs with the air programs of its member organizations.

Goals

NW-AIRQUEST is based on a strong collaboration among members aimed at:

1. Providing sound scientific advice to, and receiving appropriate input or direction from, decision-makers for the management of air quality issues within the Region;
2. Development, operation, and improvement of urban and regional numerical air quality forecast systems to form an archival database of simulated atmospheric data, including development of emission inventories and air quality observational databases for model evaluation;
3. Development, evaluation, and application of tools for specific measurement and modeling studies to address particular issues, or to answer specific science questions related to air quality in the Pacific Western North America Region;
4. Education, technology transfer, and communication to enhance understanding of current air quality issues, to improve capabilities among all members and to promote wide-spread communication of methods and results throughout the Region; and
5. Collaboration with other scientific groups, particularly organizations with similar goals in other regions, and organizations involved in the science disciplines associated with air quality, environment, and public health.

Organization and Membership

NW-AIRQUEST comprises a virtual air quality science center dependent on active involvement of all its sustaining and collaborative members.

Sustaining members encourage achievement of goals through consistent long-term collaboration and through financial support of NW-AIRQUEST products. Sustaining members have full voting capabilities and also collectively design an annual work plan for common-regional projects (i.e., projects that are designed with consideration of the common needs of regional members).

NW-AIRQUEST members strive to collectively contribute at least $110,000 per year to support work listed in the annual work plan and up to $40,000 per year for capital purchases associated with NW-AIRQUEST products. One or more members may also contribute additional funds for special projects or capital purchases in a given year. Agencies that fund these common-regional projects but are not NW-AIRQUEST members are encouraged to join NW-AIRQUEST as a sustaining member if they intend to consistently fund these projects and retain a Pacific Western North America Region focus for their involvement.

Sustaining members are expected to contribute significant funding to NW-AIRQUEST. Contributions scale with the scope of a member's organization and the member organization's size. Members are expected to contribute in line with organizations of similar scope, similar budget, and similar need for the product. Long-standing members with a proven commitment to NW-AIRQUEST may be allowed to either contribute less or not contribute at all in a particularly difficult year and still retain their sustaining membership status. University of Washington and Washington State University are sustaining members because they contribute significant in-kind services that are indispensable to the operation and improvement of our current common-regional modeling projects.

Collaborative members encourage achievement of goals through consistent long-term collaboration. They have voting privileges for all matters except those related to the funding, work plan, and access to common-regional projects. They participate in NW-AIRQUEST projects through in-kind contributions but do not fund common-regional projects. Although they strongly influence common-regional projects through collaborative and meaningful scientific input, they have no direct role in the work plan for common-regional activities and are not guaranteed access to common-regional products. In addition to in-kind contributions, it is expected that collaborative members seek financial resources to contribute to the ongoing cost of operating NW-AIRQUEST, where appropriate.

Linking Science and Policy

To help ensure that air quality policy is scientifically grounded, the various members of NW-AIRQUEST provide input to and receive direction from their respective air quality decision makers. Periodically, as appropriate, the NW-AIRQUEST co-chairs meet with policy decision-making bodies to align science and management objectives. Key components of such meetings are the discussion of the proposed annual work plan and briefings on the accomplishments and lessons learned from past work plans. The decision-making bodies in the region currently include, but are not limited to, the Northwest Air Directors Group; Tribes located in the states of Washington, Oregon, and Idaho; U.S. Federal Land Managers; and the British Columbia Visibility Coordinating Committee. Participating staff of member organizations are also expected to maintain

good two-way communication with their management regarding program science needs and policy direction, as well as NW-AIRQUEST activities, capabilities, and products.

Annual Work Plan

Sustaining and collaborative members work together to develop an annual work plan that will fulfill both short- and long-term science needs. The work plan describes two basic types of projects, reflecting different roles of NW-AIRQUEST. The first type of project is a developmental project sponsored by a member. The role of NW-AIRQUEST in a member-sponsored project is primarily advisory, facilitating collaboration and promoting information exchange and technology transfer. An example of this type of project is a modeling analysis for a State Implementation Plan revision. The second type of project is one that carries a more regional or multi-state/province support function, and is referred to in this document as common-regional projects. The role of NW-AIRQUEST in this case would include project management and implementation by sustaining members. An example of the second type of project is the operation of a regional air quality forecasting system.

Responsibilities

All sustaining and collaborative member organizations designate management- and staff-level representatives to NW-AIRQUEST that will be actively engaged in the coordination, development, and implementation of the annual work plan. Member organizations foster regular communications between their respective management- and staff-level representatives. The members readily collaborate and share expertise and resources with the other members through technology transfer and other mutually agreed upon arrangements. Sustaining member organizations seek joint funding opportunities for common-regional projects.

While supported through NW-AIRQUEST, the Laboratory for Atmospheric Research at Washington State University (LAR-WSU) will continue to maintain the virtual air quality science center and the emission inventory repository, will assist in the generation of the annual work plan, and will complete the work plan items assigned to LAR-WSU. LAR-WSU will make modeled data, emission inventory input and output data, and air quality monitoring data widely available in a format agreed to by the membership. Members with emission inventory functions are responsible for submitting their existing emission inventory data on a periodic basis to LAR-WSU. The University of Washington will continue to provide WRF weather simulations (and other products developed for the NRMC) to NW-AIRQUEST.

NW-AIRQUEST will be managed by a technical chair and supported by an administration coordinator. An EPA staff member will serve as the permanent administrative coordinator and a rotating chair will serve as the technical lead. The technical chair will rotate annually after the annual meeting in June. This technical chair

is a sustaining member; can be from any state agency, local agency, or tribe physically located within the state that will host the next annual meeting; and will be chosen by the sustaining members physically located within that state.

The administrative coordinator and technical chair will coordinate in managing the consortium and will work together to define the duties to be carried out by each party. The following lists provide an example of how the coordinator and chair might divide duties.

Administrative coordinator:

- Track members' technical needs and communicate these to the technical chair;
- Maintain member communications (e.g., maintain the listserv);
- Facilitate charter updates, when needed;
- Facilitate transitions between incoming and outgoing technical chairs;
- Present updates on NW-AIRQUEST to air quality decision makers (e.g., Northwest Air Directors and regional Tribes), in coordination with the technical chair;
- Maintain a list of voting representatives, which consists of one representative for each sustaining and collaborative member;
- Coordinate administrative planning efforts for the quarterly and annual meetings (e.g., identification of meeting space, teleconference and audio lines, distribution of agendas; development and distribution of meeting minutes); and
- Perform outreach, as appropriate, to potential new NW-AIRQUEST members, in coordination with the technical chair.

Technical chair:

- Develop quarterly and annual meeting agendas;
- Lead development of the annual work plan;
- Facilitate the quarterly and annual meetings;
- Present updates on NW-AIRQUEST to air quality decision makers (e.g., Northwest Air Directors and regional Tribes), in coordination with the administrative coordinator;
- Lead the voting process for members' proposals; and
- Perform outreach, as appropriate, to potential new NW-AIRQUEST members, in coordination with the administrative coordinator.

Operations

For each project undertaken by NW-AIRQUEST, there will be an "executive sponsor" from one of the sustaining members. This person will be responsible for ensuring NW-AIRQUEST's project goals are carried out and will be responsible for reporting back results and updates to NW-AIRQUEST if none come from the project principal investigator.

The group will generally operate through consensus decision-making for common-

regional projects and administrative matters. In the event that consensus cannot be achieved or that it is most practical to collect opinions via a voting process, any sustaining or collaborative member may call for a vote at any time on any issue, providing a quorum consisting of three-fifths of the total number of regularly participating voting member organizations participates in the vote. At least half of the members that vote must vote for a given proposal in order for a proposal to pass.

The administrative coordinator will maintain a list of one voting representative per sustaining and collaborative member organization. The consortium will strive to make decisions on proposed approaches during the course of meetings and will elevate more complex issues to the voting representatives on an as-needed basis. Each voting representative will be selected by their respective organization to represent the organization's viewpoints on both technical and policy matters. Each voting representative is responsible for briefing their organization on the proposal to be voted on and achieving any necessary approvals within their organization before submitting their vote.

While the EPA is considered a sustaining member of NW-AIRQUEST and serves as the permanent administrative coordinator, the EPA does not have a role in defining the technical work of the consortium. Therefore, EPA participants may offer technical advice and insight but will not advocate for particular technical approaches or have voting privileges. Rather, the EPA will assist in collecting the viewpoints and votes of other members and communicating this information to the technical chair, other consortium members, and air quality decision makers (e.g., the Northwest Air Directors and regional Tribes).

Publications will not appear under the name of the NW-AIRQUEST unless a unanimous decision is made to do so.

Meetings

At least three meetings will be held annually: a face-to-face business and science meeting, and two science meetings. The purpose of the business meeting is to finalize the annual work plan. The science meetings will primarily be an opportunity for peer review, evaluation and collaboration, and to assess long-term scientific direction.

Active and Expiry Date

This instrument is executed as of the last signature date shown below. At five year intervals from that date, the charter will be subject to review, and will be revised if necessary.

Non-Fund Obligating Document

This instrument is neither a fiscal nor a funds obligation document. Any endeavor involving reimbursement or contribution of funds between the parties to this instrument will be handled in accordance with applicable laws, regulations, and procedures including those for Government procurement and printing. Such endeavors will be outlined in separate agreements that shall be made in writing by representatives of the parties and shall be independently authorized by appropriate statutory authority. This instrument does not provide such authority. Specifically, this instrument does not establish authority for noncompetitive award to the cooperator of any contract or other agreement. Any contract or agreement for training or other services must fully comply with all applicable requirements for competition.

Additional Terms Concerning EPA Participation in the Consortium

All commitments made by EPA in this Charter are subject to the availability of appropriated funds. Nothing in this Charter, in and of itself, obligates EPA to expend appropriations or to enter into any contract, assistance agreement, interagency agreement, or incur other financial obligations that would be inconsistent with Agency budget priorities. All parties to this Charter waive any claim for compensation for services rendered to EPA in connection with any activities it carries out in furtherance of this Charter. This Charter does not exempt parties to the Charter from EPA policies governing competition for assistance agreements. Any transaction involving reimbursement or contribution of funds between EPA and the parties to this Charter will be handled in accordance with applicable laws, regulations, and procedures under separate written agreements.

Pursuant to Federal ethics rules, EPA employees may not, with limited exceptions, endorse or promote products or services offered by or provided by any non-federal entity. Nothing in this Charter constitutes an endorsement by EPA to any products or services, or any fundraising activity or promotion relating to other parties to the Charter. All parties to the Charter agree not to make statements to the public at workshops and meetings, promotional literature, on its web site or through any other media that imply that EPA or any of its employees endorses any commercial service or product. In addition, the parties to the Charter agrees not to make statements that imply that EPA supports any party to the Charter's efforts to raise public or private funds. Any statements or promotional materials that describe this Charter must be approved in advance by EPA and the sustaining membership.

Membership as of May 2020

The **Sustaining Members** are:

Federal
- U.S. Environmental Protection Agency, Region 10

Tribes in Washington, Oregon, and Idaho, including
- Nez Perce Tribe
- Swinomish Indian Tribal Community

State and Local Agencies
- Idaho Department of Environmental Quality
- Oregon Department of Environmental Quality
- Oregon Local Air Agencies
  - Lane Regional Air Protection Agency
- Washington Department of Ecology
- Washington Local Air Agencies
  - Benton Clean Air Agency
  - Northwest Clean Air Agency
  - Olympic Regional Clean Air Agency
  - Puget Sound Clean Air Agency
  - Southwest Clean Air Agency
  - Spokane Regional Clean Air Agency
  - Yakima Regional Clean Air Agency

Institutions
- University of Washington
- Washington State University

The **Collaborative Members** are:

International

Environment and Climate Change Canada (ECCC)
Ministry of Environment & Climate Change Strategy (ENV)

Federal

USDA Forest Service, Pacific Northwest Research Station

Signatures

The following signatures constitute agreement to participate in the roles defined above:

Sustaining Members

KRISHNASWAMY VISWANATHAN Digitally signed by KRISHNASWAMY VISWANATHAN Date: 2020.05.29 13:56:18 -07'00'

Krishna Viswanathan Date
Director
Air and Radiation Division
US Environmental Protection Agency
Region 10
Seattle, WA
USA

Samuel N. Penney 27 May 2021

Samuel N. Penney Date
Chairman
Nez Perce Tribal Executive Committee

Tiffany Floyd 05/20/2020

Tiffany Floyd Date
Air Quality Division Administrator
Idaho Department of Environmental Quality
Boise, ID
USA

6/23/2020

Ali Mirzakhalili Date
Air Quality Division Administrator
Oregon Department of Environmental Quality
Portland, OR
USA

Merlyn L. Hough June 26, 2020

Merlyn Hough Date
Director
Lane Regional Air Protection Agency
Springfield, OR
USA

June 1, 2020

Kathy Taylor Date
Air Quality Program Manager
Washington Department of Ecology
Olympia, WA
USA

Sustaining Members

June 5, 2020

Robin Priddy
Robin Priddy — Date
Executive Director
Benton Clean Air Agency
Kennewick, WA
USA

June 3, 2020
Mark Buford — Date
Executive Director
Northwest Clean Air Agency
Mount Vernon, WA
USA

June 3,,2020
Fran McNair — Date
Executive Director
Olympic Regional Clean Air Agency
Olympia, WA
USA

May 19, 2020
Craig Kenworthy — Date
Executive Director
Puget Sound Clean Air Agency
Seattle, WA
USA

June 9, 2020
Uri Papish — Date
Executive Director
Southwest Clean Air Agency
Vancouver, WA
USA

June 3, 2020
Julie Oliver — Date
Executive Director
Spokane Regional Clean Air Agency
Spokane, WA
USA

Sustaining Members

Keith M. Hurley June 10, 2020

Keith Hurley Date
Executive Director
Yakima Regional Clean Air Agency
Yakima, WA
USA

06/15/2020

Christopher Keane Date
Vice President of Research
Washington State University
Pullman, WA
USA

Clifford Mass 6/12/2020

Clifford F. Mass Date
Professor
Department of Atmospheric Sciences
University of Washington
Seattle, WA
USA

Steve Edwards 7.16.20

Steve Edwards Date
Chairman
Swinomish Indian Tribal Community

Collaborative Members

Ouellet, Mario

Digitally signed by: Ouellet, Mario
DN: CN = Ouellet, Mario C = CA O = GC OU = EC-EC
Date: 2020.08.26 18:36:17 -06'00'

Mario Ouellet Date
Regional Director
Prediction Services Operations- West
Environment and Climate Change Canada
Vancouver, BC
Canada

PAUL ANDERSON

Digitally signed by PAUL ANDERSON
Date: 2020.07.31 12:10:01 -07'00'

Paul Anderson Date
Station Director (Acting)
USDA Forest Service
Pacific Northwest Research Station
Corvallis, OR
USA

August 7, 2020

Christa Zacharias-Homer Date
Director, Environmental Standards
Environmental Protection Division
Ministry of Environment & Climate Change Strategy
Victoria, BC
Canada